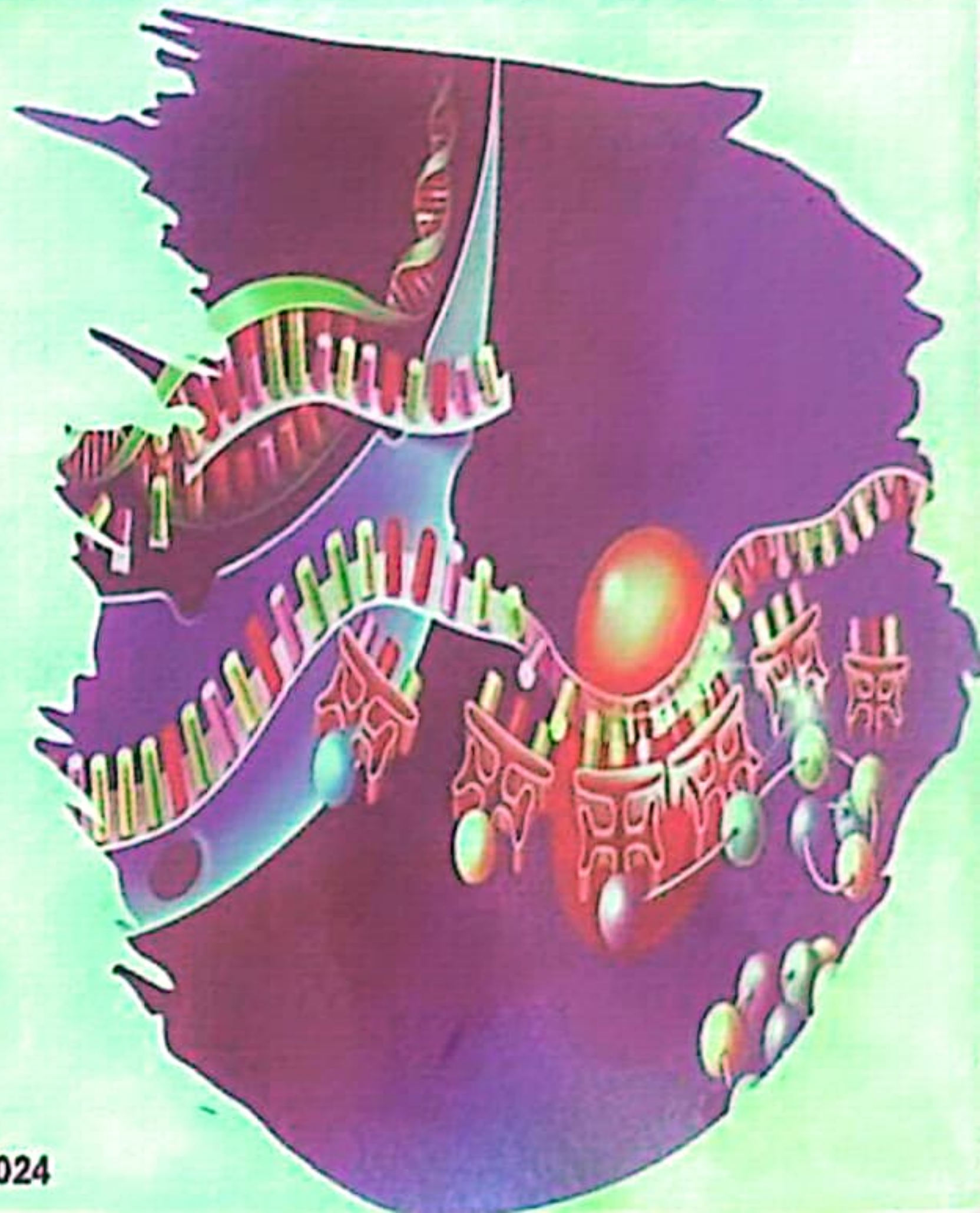


Lectuer 8

# Genetic Expression

- Prof. Dr. Walaa Slouma  
Professor of Genetics

23, 25/11/2024





# Outlines

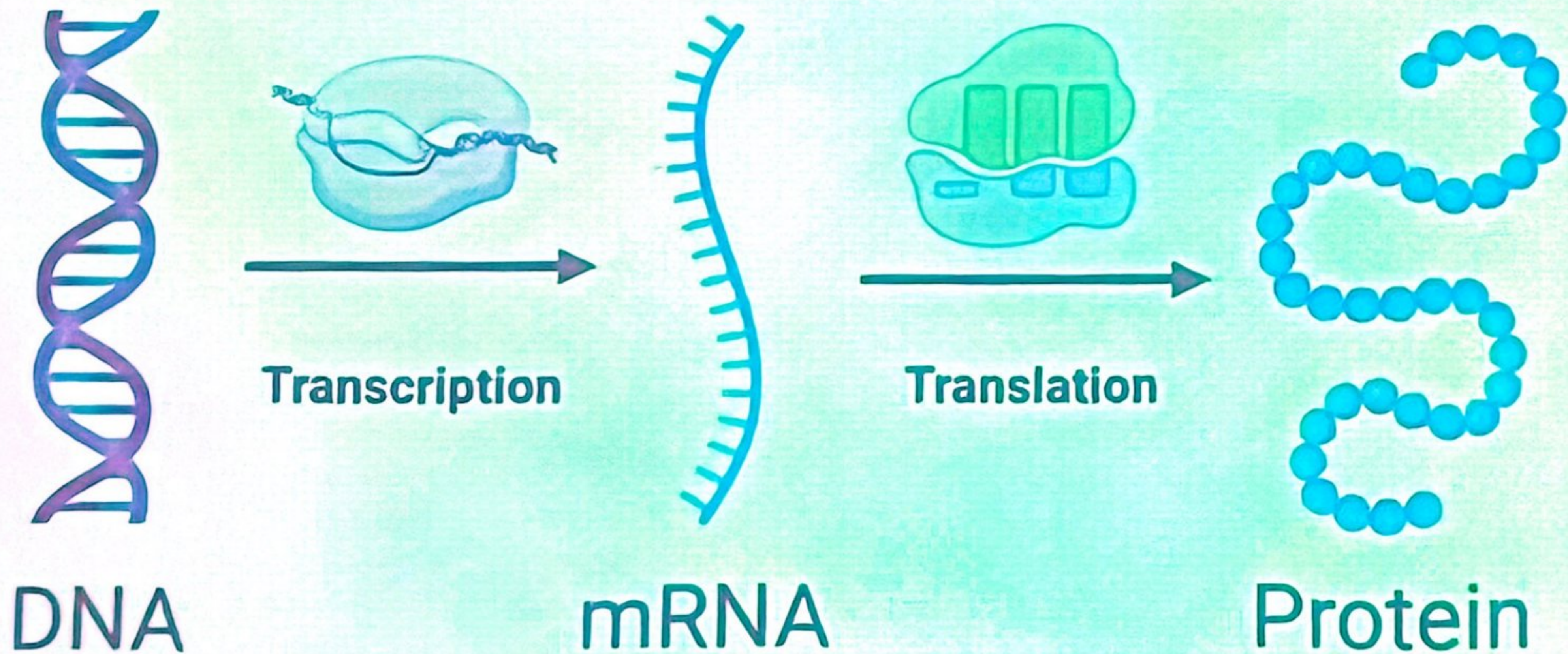
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- **Translation: Protein synthesis**
  - Ribosome structure
  - Transfer RNA structure
  - Phases of Translation
- **Regulation of protein synthesis**
- **Mutation**



# Genetic Expression





# Translation



is the conversion of genetic information transcribed on a *messenger RNA* to a sequence of *amino acids* in a polypeptide chain

**Q.** How do triplet of ribonucleotides present in *mRNA* can direct *specific a.a* into their correct places along polypeptide chain??



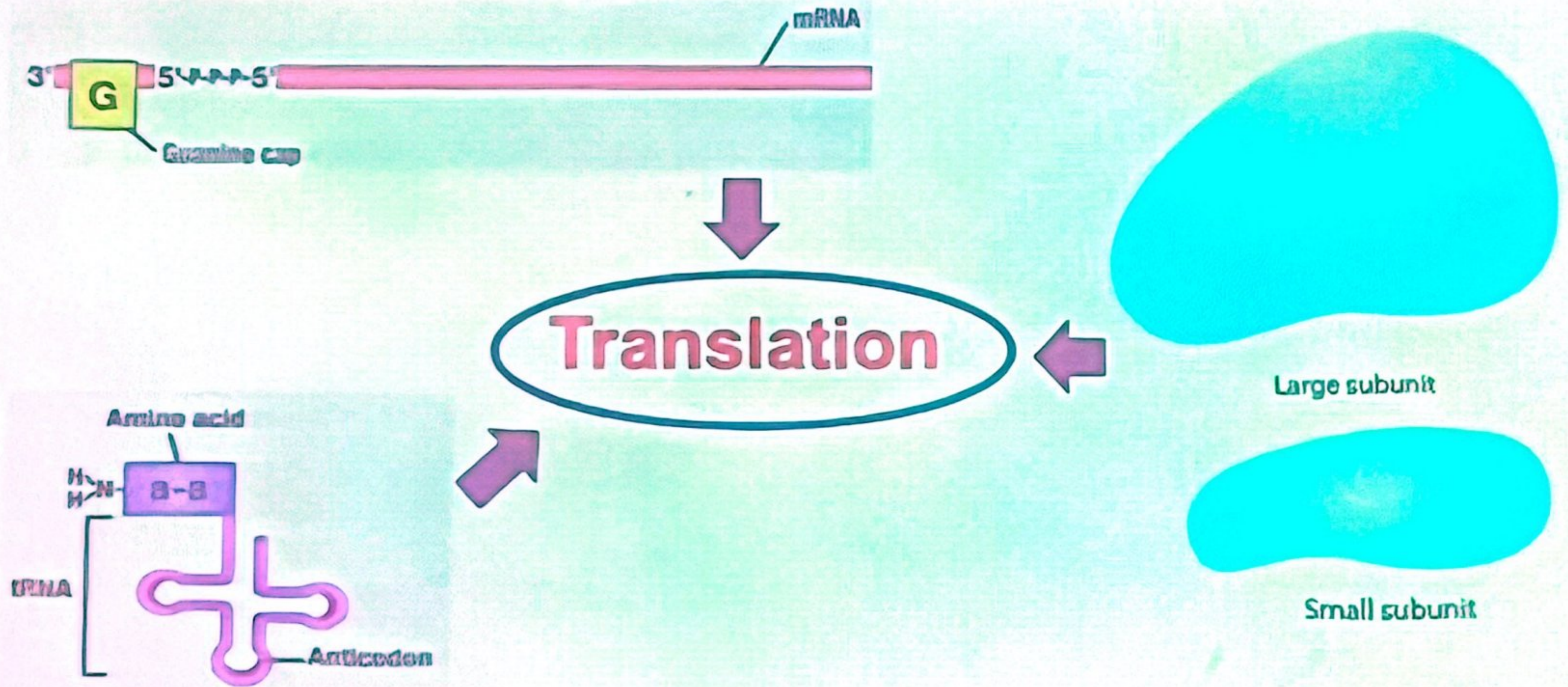
## Transfer RNA (tRNA)

(adaptor molecule)

### Codon- anticodon Recognition



# Translation





# Ribosome structure



- **The ribosome** plays an essential role in the expression of genetic information (the sites of protein synthesis)



# Ribosome structure



- **The ribosome** plays an essential role in the expression of genetic information (the sites of protein synthesis)

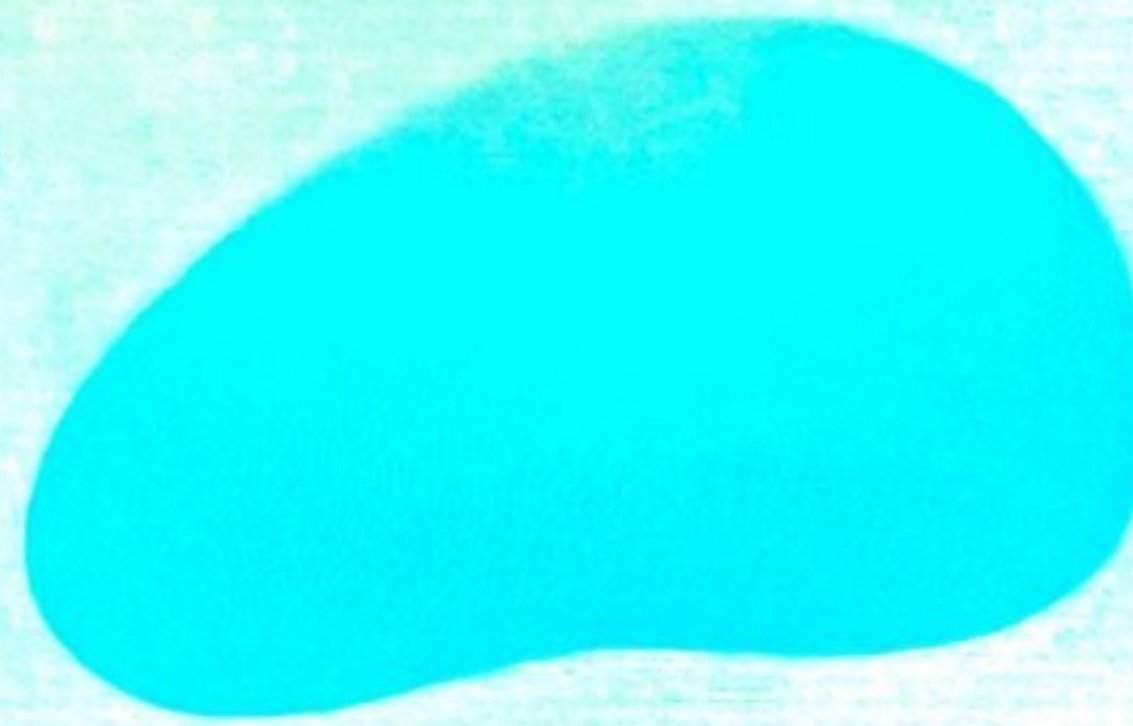
A bacterial cell  
contains about *10,000*  
ribosomes

Eukaryotic cells  
contain many times more than  
this amount of ribosomes

- **The ribosome** consists of a *small subunit* and a *large subunit*



Small subunit



Large subunit



# Ribosome structure



| Monosome   |         | Prokaryotes<br>70S                                         |                            | Eukaryotes<br>80S                                                                  |                            |
|------------|---------|------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------|----------------------------|
|            |         | Large subunit (50S)                                        | Small subunit (30S)        | Large subunit (40S)                                                                | Small subunit (60S)        |
| Components | RNA     | <b>23S rRNA</b><br>3000 nt<br><br><b>5S rRNA</b><br>120 nt | <b>16S rRNA</b><br>1500 nt | <b>28S rRNA</b><br>5000 nt<br><br><b>5S rRNA</b><br>120 nt<br><br><b>5.8S rRNA</b> | <b>18S rRNA</b><br>2000 nt |
|            | Protein | <b>34 proteins</b>                                         | <b>21 proteins</b>         | <b>40 proteins</b>                                                                 | <b>30 proteins</b>         |

Gene coding for rRNA are *Redundant*

- In **euokaryotes**, many more copies of rRNA present, varies between species and present in cluster at various chromosomal sites
- In **E.coli**, **4** copies of a single DNA seq. which codes for all **3** rRNA



## Prokaryotes

30S { 16S rRNA  
21 Proteins

50S { 23S rRNA  
5S rRNA  
34 Proteins

70S

small  
subunit

large  
subunit

complete  
ribosome

## Eukaryotes

18S rRNA  
30 Proteins } 40S

28S rRNA  
5.8S rRNA  
5S rRNA  
40 Proteins } 60S

80S



# Transfer RNA structure



- **Smallest** class of RNA and **most stable**
- Composed of 75 to 90 nt.
- Have the **same structure** in **prokaryotes** and **eukaryotes**

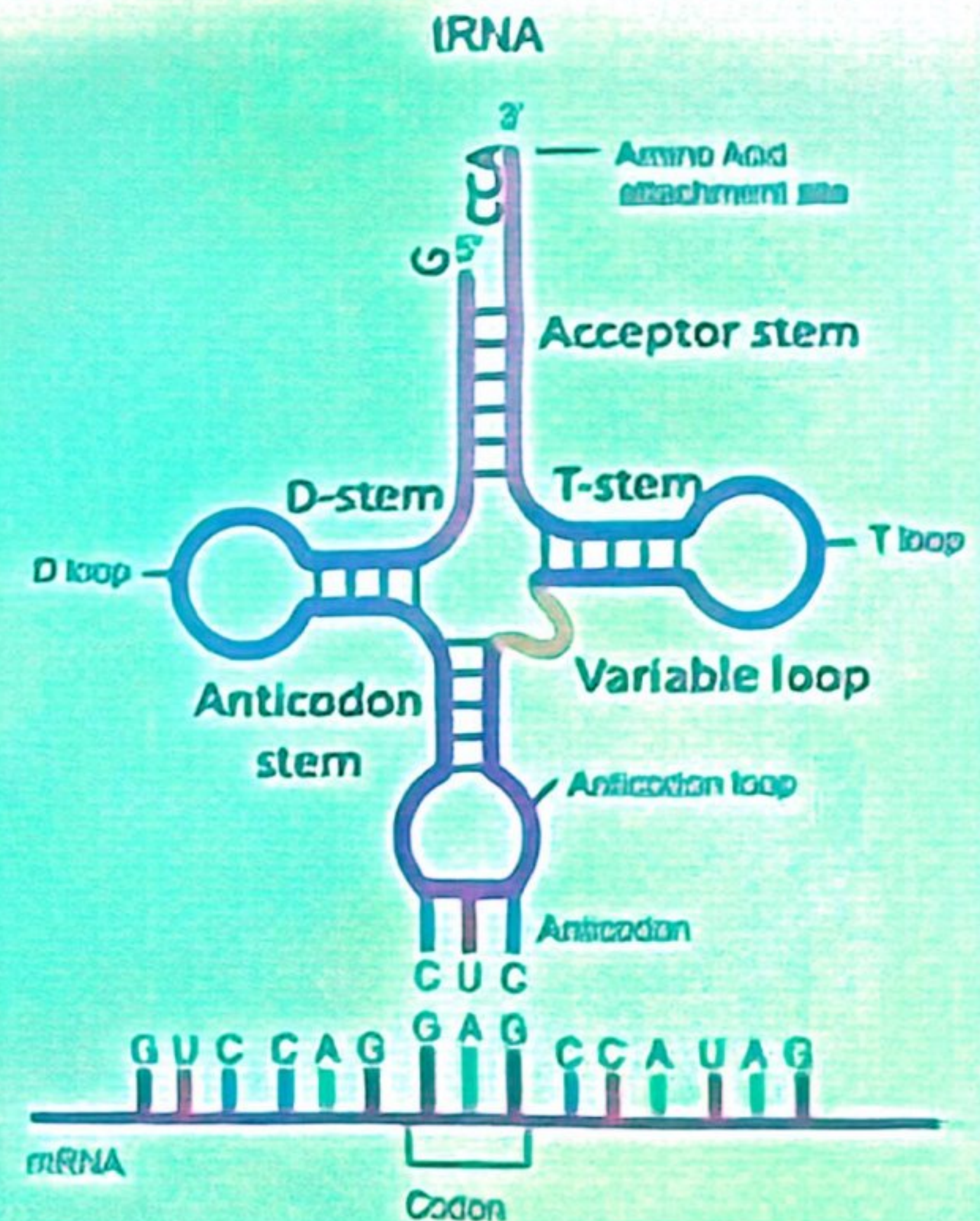
▪ Pre-tRNA is large  $\xrightarrow{\text{processed}}$  small size tRNA

ex. pre-tRNA<sup>tyr</sup> (126 nt)  $\longrightarrow$  tRNA<sup>tyr</sup> (77 nt)

- **Holley and his coworkers (1965)**

Two-dimensional **cloverleaf** model acc.to paired stems and **unpaired loops**

- Loops consistently contain **modified base** (**Inosin**, **Methyl inosin**, **Methyl guanosine**)





# Phases of Translation



**Translation may be subdivided into four phases:**

**1- Charging of tRNA**

**2- Polypeptide Chain Initiation**

**3- Polypeptide Chain Elongation**

**4- Polypeptide Chain Termination**



# 1- Charging of tRNA



- The tRNA chemically linked to their respective amino acids, this under the direction of specific enzyme called (**aminoacyl synthetase**)

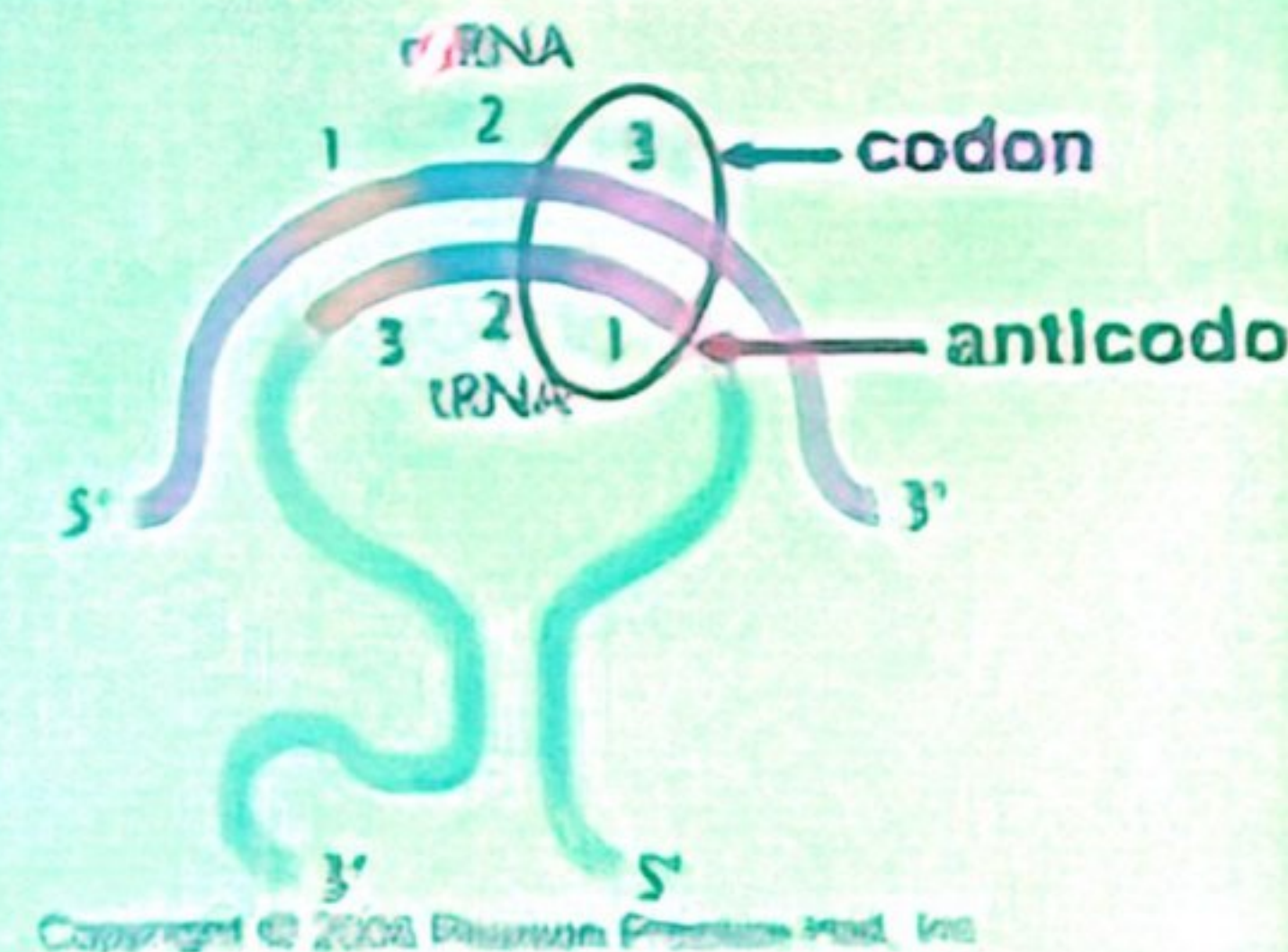
20 a.a  $\longrightarrow$  at least 20 different tRNA

**But** 61 triple codes and this indicated theoretically to presence of 61 tRNA and enzyme

However, due to **Wobbling**

| First member in anticodon | 3 <sup>rd</sup> position in codon |
|---------------------------|-----------------------------------|
| U                         | A or G                            |
| G                         | C or U                            |
| I                         | C or U or A                       |

So, there is 30 tRNA and 30 corresponding enzyme in cytoplasm





**Amino acid (aa<sup>x</sup>) + ATP (aminoacyladenylate (aa-AMP))**  
**+ Aminoacyl synthetase<sup>x</sup>**

**aa<sup>x</sup>-AMP-enzyme complex + ppi**

↓ tRNA\*

**tRNA<sup>x</sup>.aa<sup>x</sup>** + **Aminoacyl synthetase<sup>x</sup>**  
(aminoacyl tRNA)



## 2- Polypeptide Chain Initiation



For the **initiation of translation** in *E. coli*, the following components have to be present:

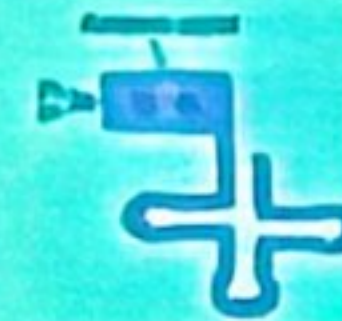
1. The mRNA molecule



2. Large and small ribosome subunits



3. A specific *initiator tRNA* molecule



4. GTP (guanosine triphosphate)

5.  $Mg^{++}$

6. At least three protein *initiation factors* (IF1, IF2, IF3)



# Various protein factors involved during translation

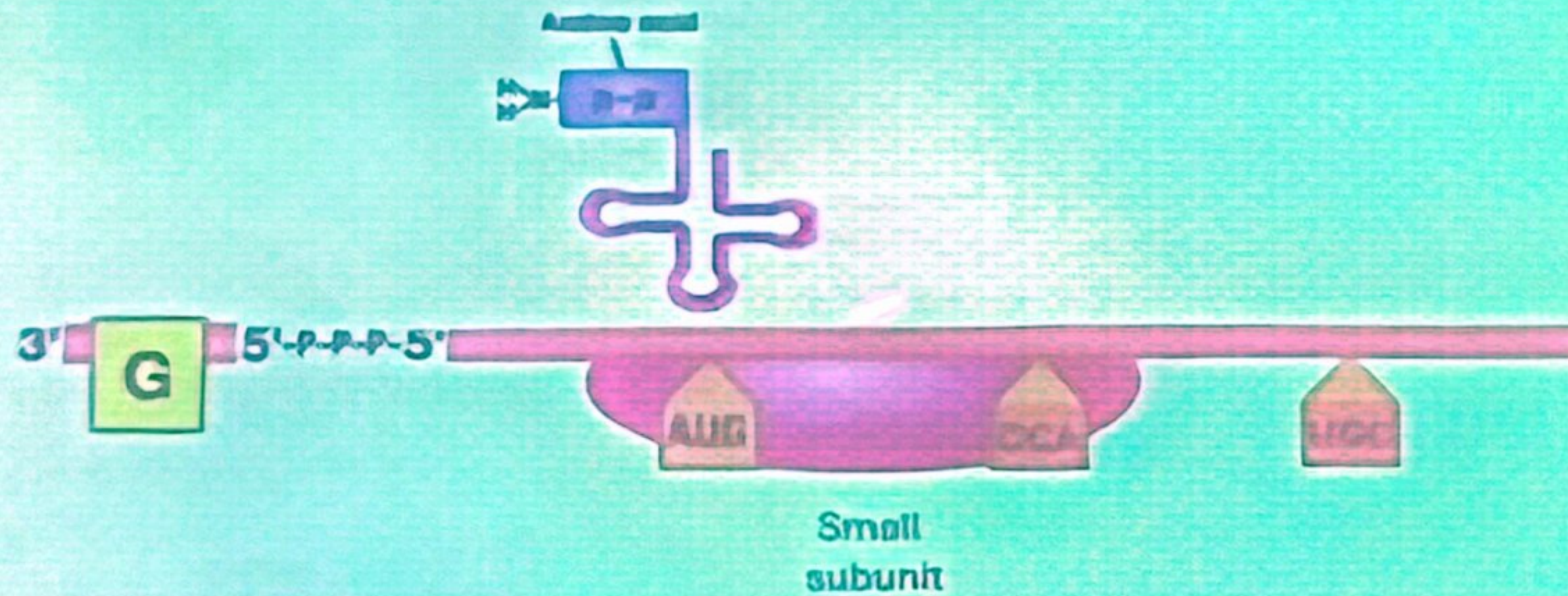
| Process                                               | Factors | Role                                                                                                                                             |
|-------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Initiation of translation                             | IF1     | Stabilizes 30S subunit                                                                                                                           |
|                                                       | IF2     | Binds <del>Phe</del> met-tRNA to 30S-mRNA complex                                                                                                |
|                                                       | IF3     | Binds 30S subunits to mRNA; dissociates monosomes into subunits following termination                                                            |
| Elongation of polypeptides                            | EF-Tu   | Binds Gtp; brings aminoacyl-tRNA to the A site of ribosome.                                                                                      |
|                                                       | EF-Ts   | Generates active EF-Tu                                                                                                                           |
|                                                       | EF-G    | Stimulates translocations; Gtp-dependant                                                                                                         |
| Termination of translation and release of polypeptide | RF1     | Catalyzes release of the polypeptide chain from tRNA and dissociating of the translocation complex; specific for UAA and UAG termination codons. |
|                                                       | RF2     | Behaves like RF1; specific for UGA and UAA codons                                                                                                |
|                                                       | RF3     | Stimulate RF1 and EF2                                                                                                                            |



## 2- Polypeptide Chain Initiation

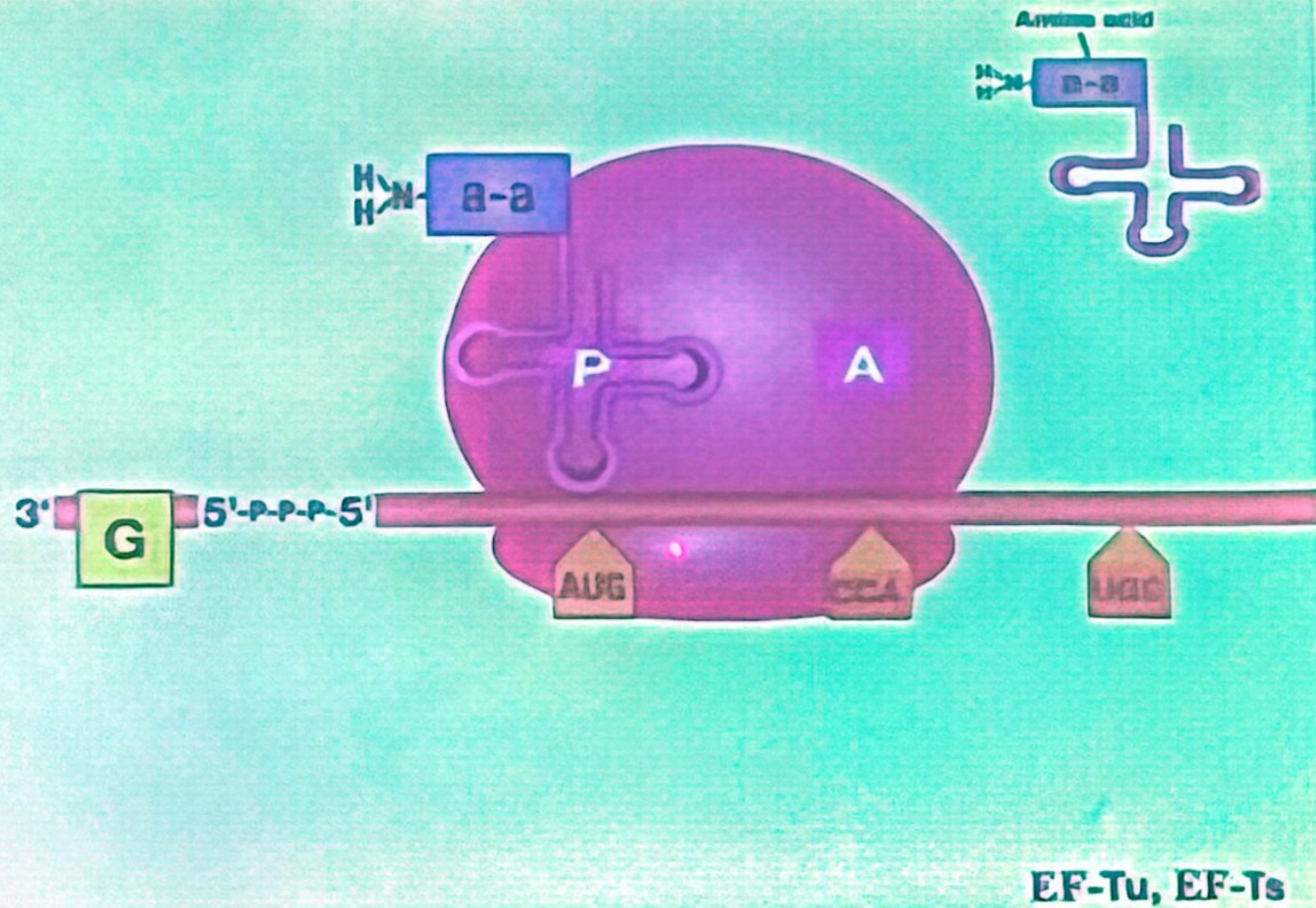


### Initiation Complex



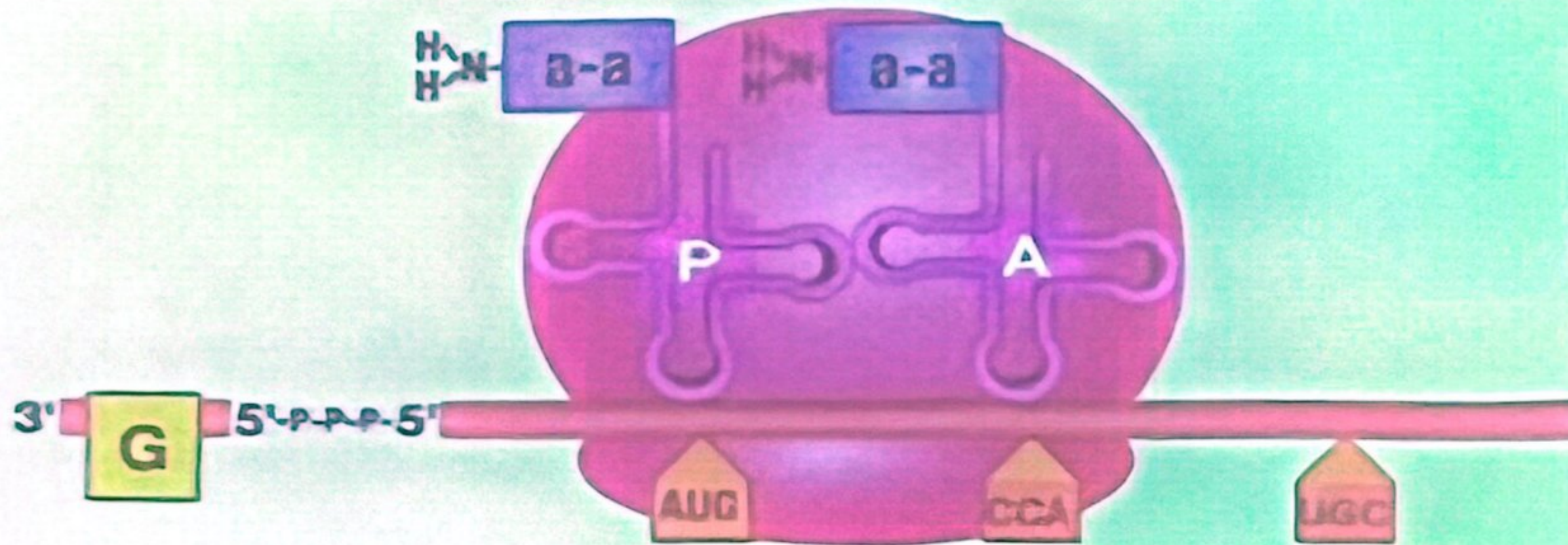


## 2- Polypeptide Chain Initiation



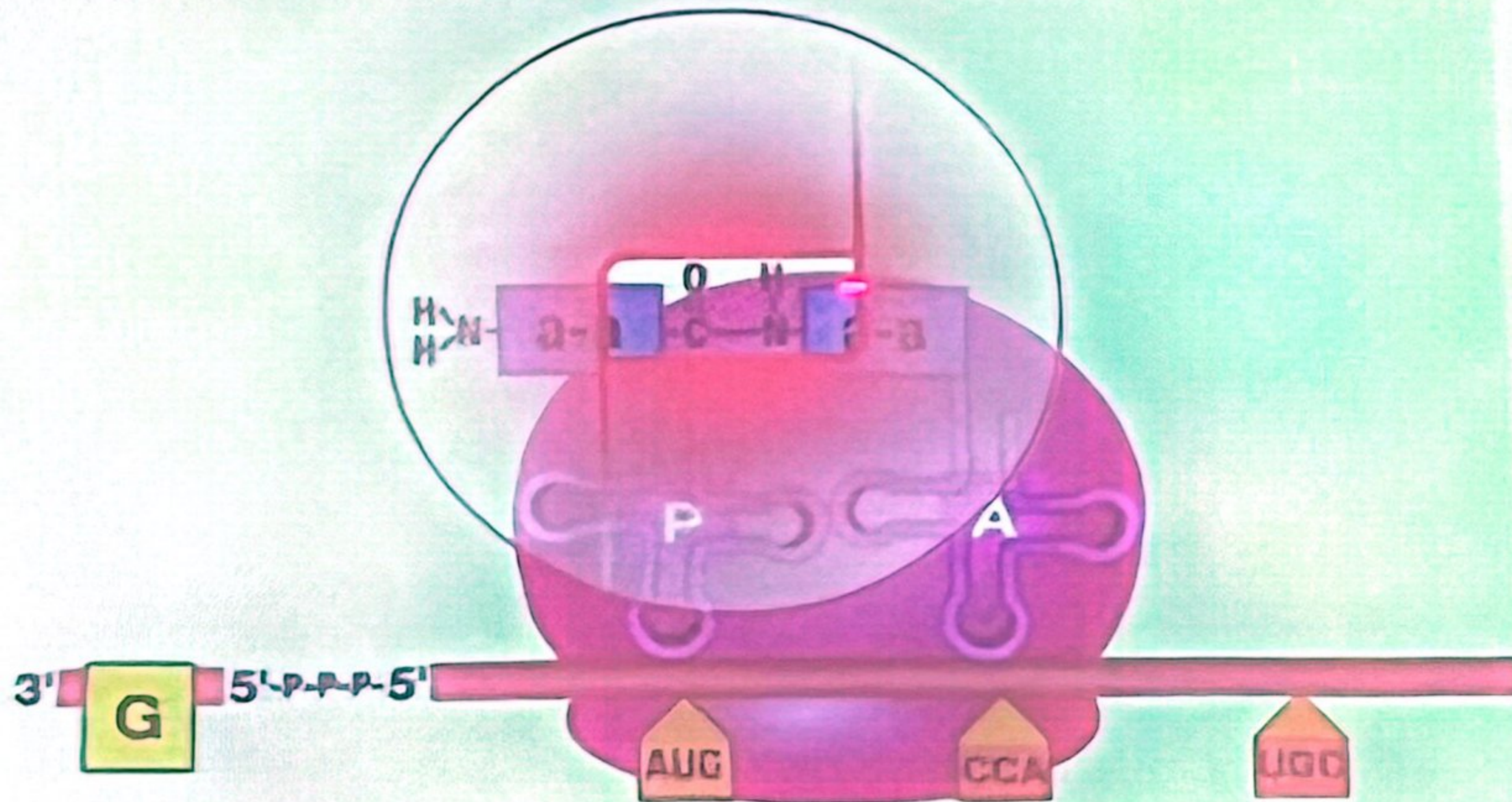


### 3- Polypeptide Chain Elongation





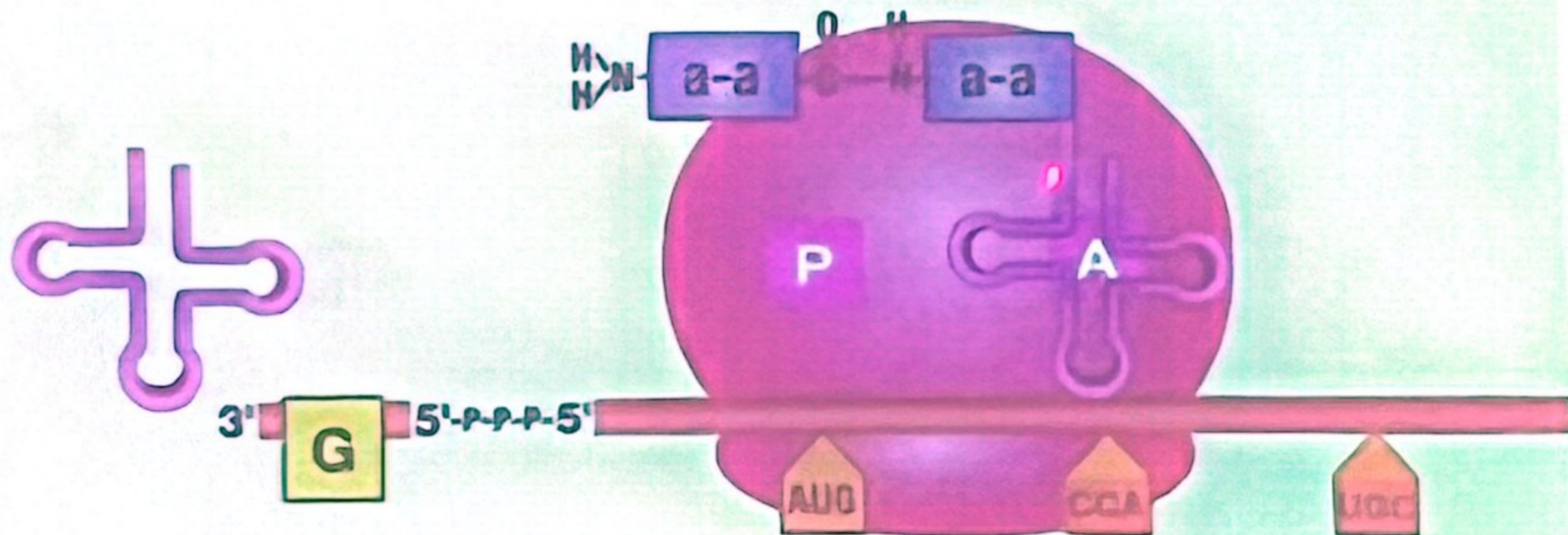
### 3- Polypeptide Chain Elongation



Dr. Da Muta

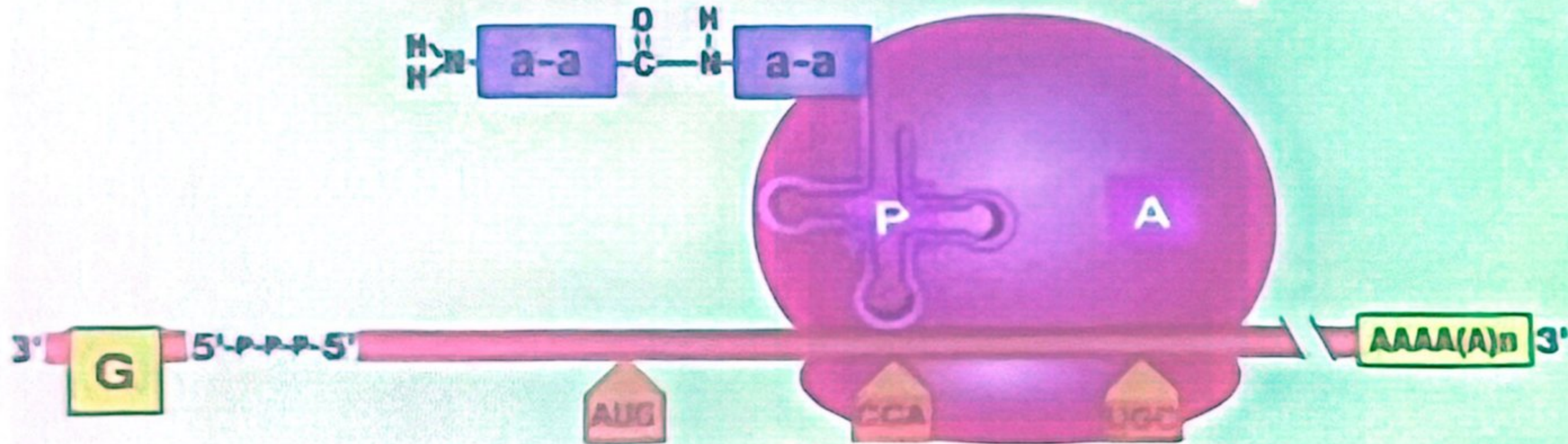


### 3- Polypeptide Chain Elongation





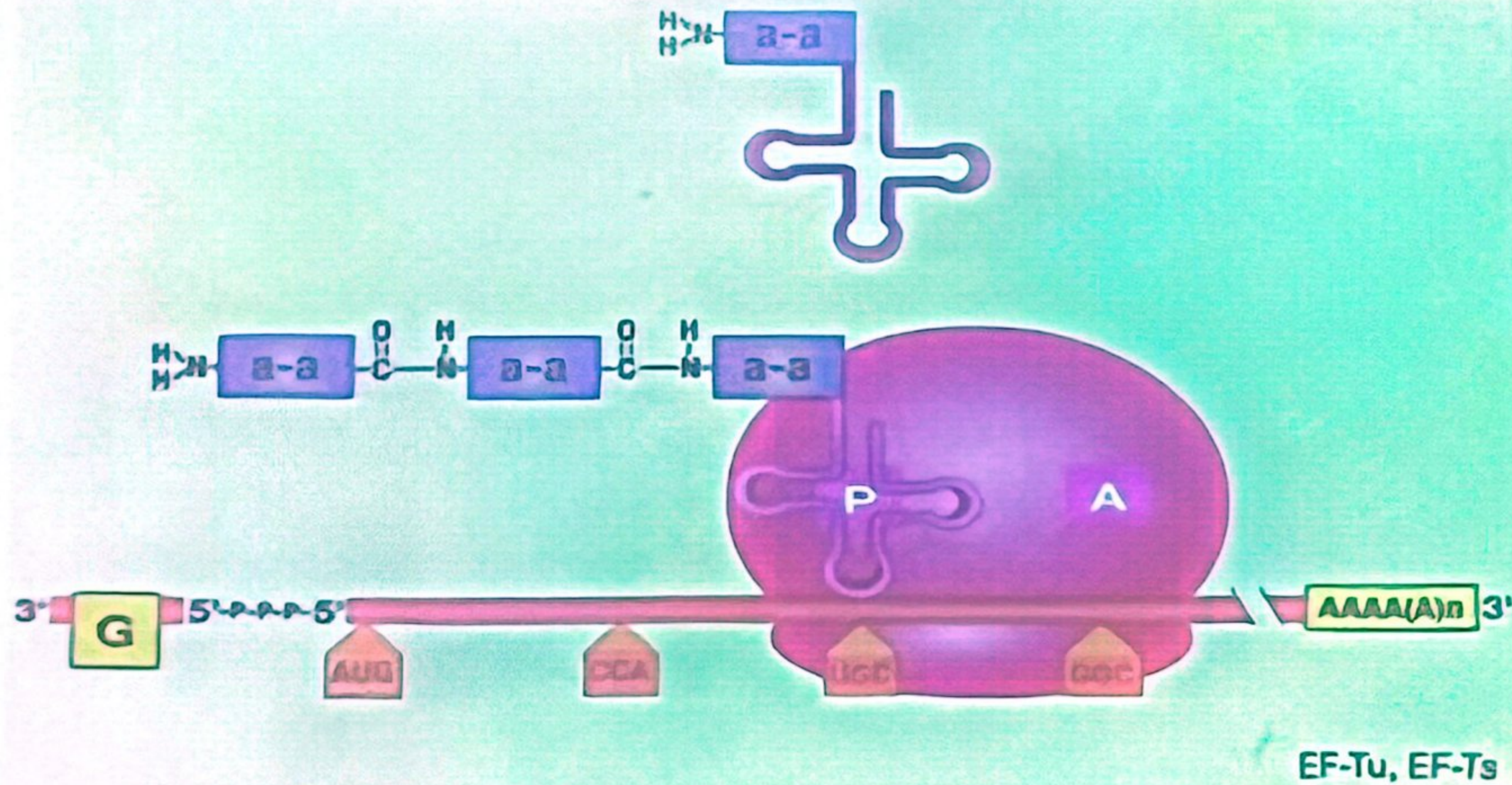
### 3- Polypeptide Chain Elongation



Translocation. (EF-G), GTP

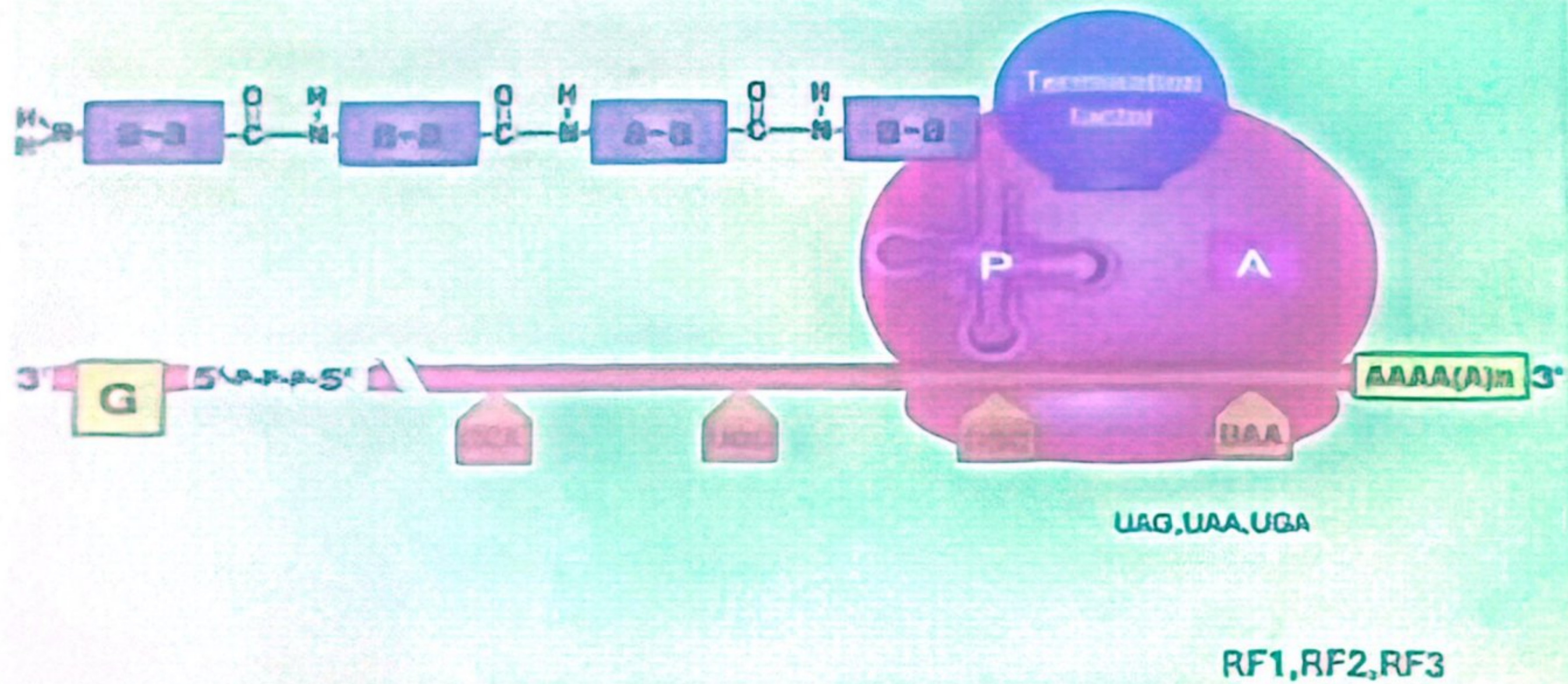


### 3- Polypeptide Chain Elongation



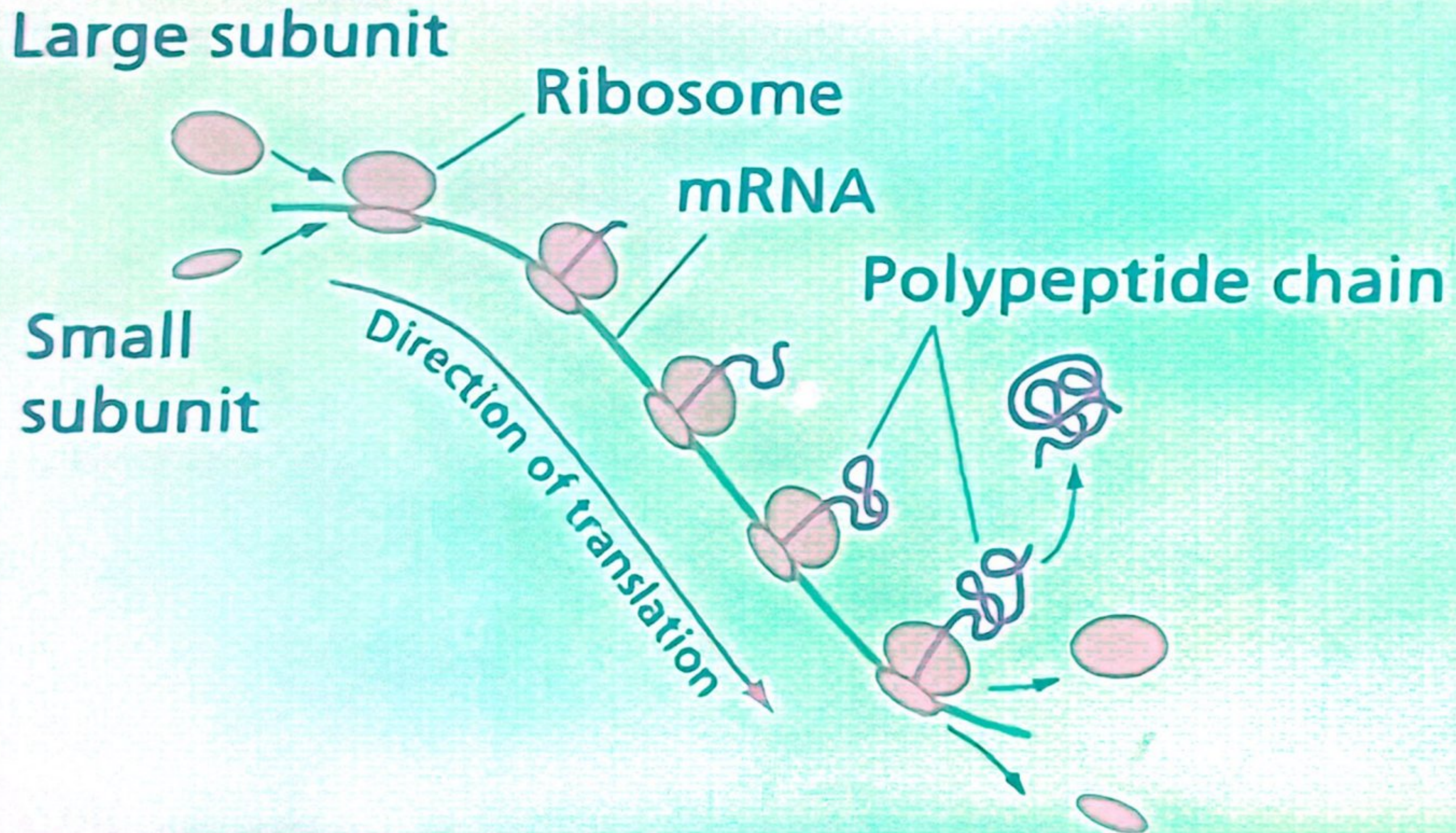


### 3- Polypeptide Chain Elongation



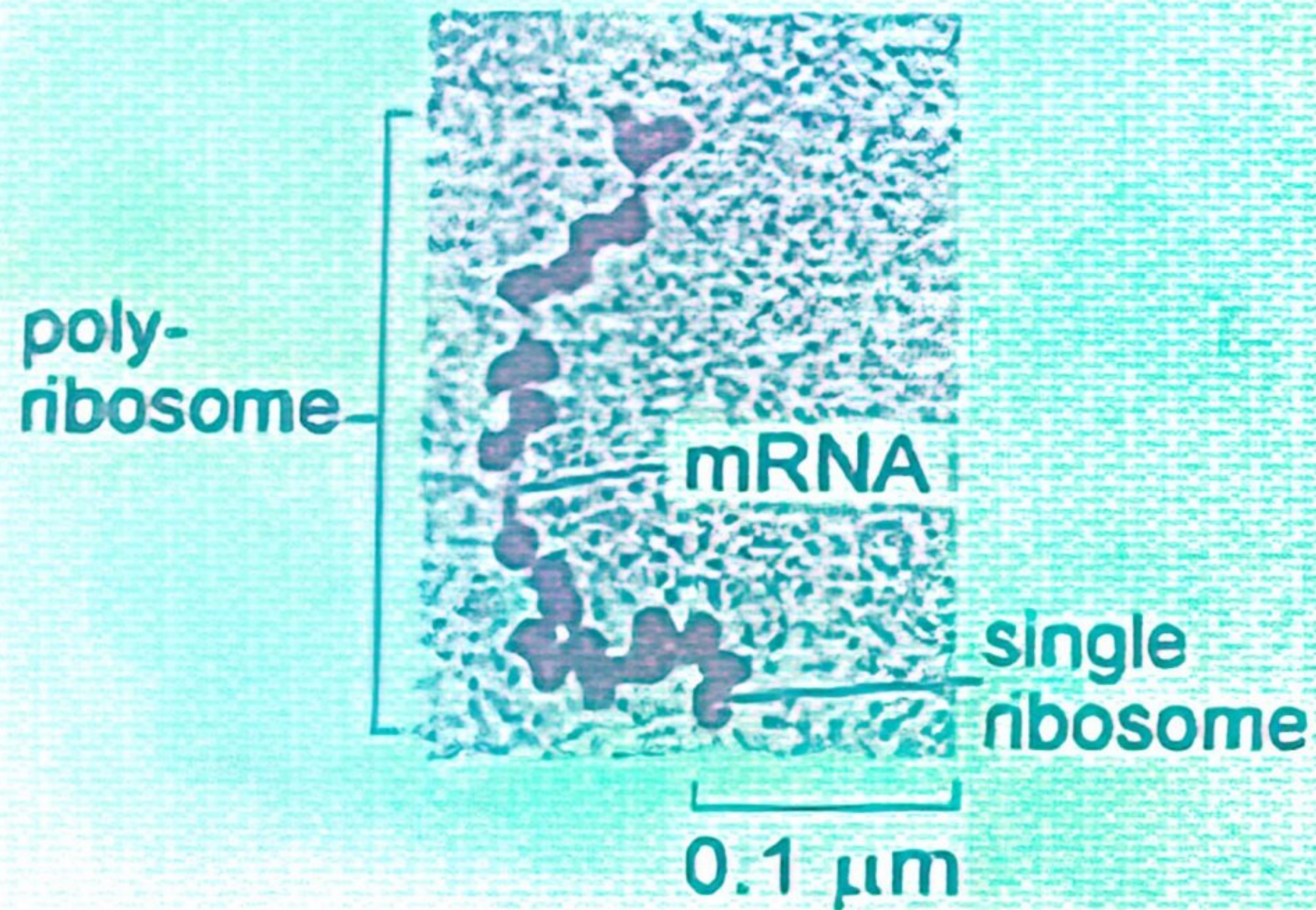


# Polyribosomes or Polysomes





# Polyribosomes or Polysomes





# Regulation of protein synthesis

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**Genetic Studies** —→ Not all genes present in a cell are active in directing specific process of protein synthesis

- ✓ This **variable** gene **activity** in different timings indicate a regulatory mechanisms for protein synthesis

**So,** The genes can be switched **on** or **off** depending upon:

1. Position of a cell in the body
2. Stage of body development
3. The external environment



# Regulation of protein synthesis



Genes in the body divided into two types:

## Constitutive genes

-These genes are expressed **continuously**, and their products are found in the cell in almost the same amount at all the time

- Controls governing the expression of it are **invariant**

ex. Enzyme of **glycolysis**

## Regulated genes

- On call, are expressed according to body needed so amount of its product differ from time to time

- Its expression are **modified** by the presence or absence of chemical stimuli in the gene's environment

ex.

- Enzyme needed in the pathway for **synthesis of a.a**
- Enzymes used in metabolism of **Lactose, arabinose, galactose**



# Mutation

**Dr. Walaa Slouma**  
Professor of Genetic



# Outlines

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- **Mutations**
- **Classification of Mutations**



# Mutations



- Failure to store the genetic information faithfully, It will affect its expression and propagated by replication
- In the past, the term mutation included chromosomal aberrations. Now it is often reserved for alteration of information within a single gene

**Point mutation** —→ is more specific since the change usually involves a single nucleotide change



# Classification of Mutations:



## a. Origin

1. Spontaneous mutation
2. Induced mutation

## b. Cell type

1. Somatic mutation
2. Gametic mutation

## c. Type of change

1. Substitutional mutation
2. Frameshift mutation



# a. Origin



## 1. Spontaneous mutation

- These are considered as arising in nature (arise by chance)



No specific factors, other than natural elements, are associated with their occurrence

- it is considered as Random errors in the storage of genetic information
- Causes: exposure to background radiation (cosmic rays or radioactive minerals), chemical and thermal elements



Result in error during replication

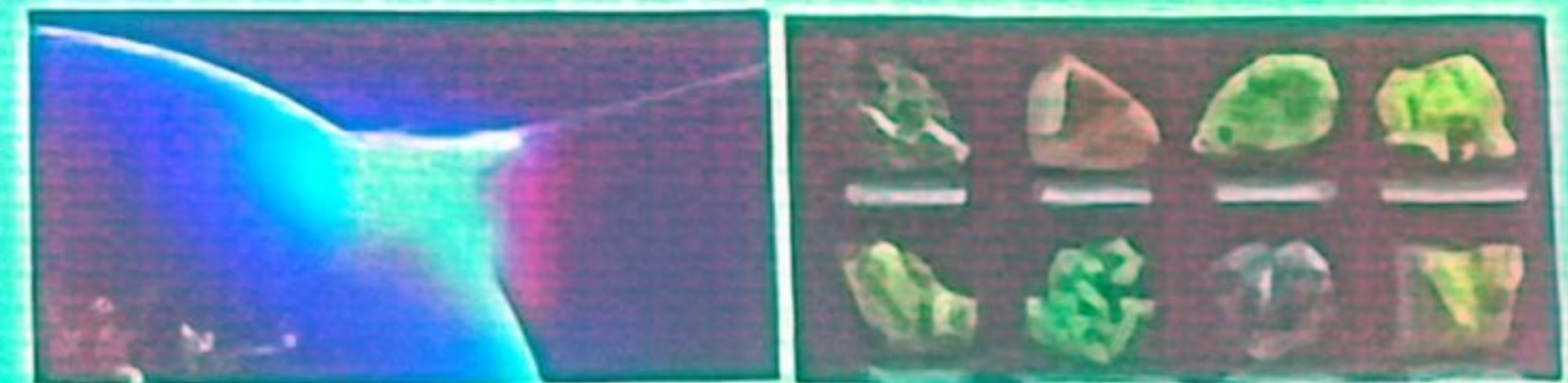


DNA polymerases make rare errors



an incorrect base is inserted

Error in genetic code → Change amino acid composition





# 1. Spontaneous mutation

- **Spontaneous mutation rate** → Has been estimated to average about **once in 100 millions ( $10^{-8}$ )** cell divisions.
- Mutation rate is higher in **Drosophila** and humans ( $10^{-6}$  to  $10^{-5}$ ) per gamete formed.
- The reason for this **large variation** in spontaneous mutation rate is **not clear**



reflect **relative efficiency** of enzyme systems responsible for repair errors created during replication